Delphes for HH→ $b\bar{b}\tau\tau$ NSBI:
full production tuned and validated against CMS

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Abstract

The full Delphes production is tuned, ntuplised and validated: 3×5M signal events at $\kappa_\lambda = 0, 1, 5$ and 299M $t\bar{t}$ events, 167 GB of analysis ntuples. In the semi-leptonic channel — the one the CMS NSBI result uses — Delphes now reproduces CMS to within a few percent in yield (30 491 vs 28 181 at $\kappa_\lambda = 0$) and agrees in shape on m_{HH}, m_{bb}, p_T^{HH}, ΔR_{bb}, $\Delta\phi_{HH}$, $p_T^{H1,H2}$, $\cos\theta^*$, both τ p_T and p_T^{miss}. Two things were found in the course of this validation and are worth the PI's attention: the CMS lepton collections were being read *without* ID or isolation, which both broke the semi-leptonic comparison and biased the lepton scale factors (Sec. 2.1); and the b-jet energy *resolution* is untuned by construction, only its scale (Sec. 3.3). One physics residual remains: Delphes makes too many wide-angle τ pairs, visible as an excess at $\Delta R_{\tau\tau} \simeq 2.5$–$3.2$ and as high tails on $m_{\tau\tau}$ and m_{vis} (Sec. 3.1). Since the last note the $t\bar{t}$ background has also been compared against its own CMS anchor for the first time and the correction works (Sec. 2.3), and one new defect was found: FastMTT returns no solution for the majority of events, because a Delphes τ_h carries a jet mass far above m_τ (Sec. 3.2).

1 Where we stand

1.1 Production

Everything downstream of Delphes: samples are never re-made, and every correction is a map measured on a private CMS 2024 NanoAODv15 anchor and transferred. The anchor is the `PowhegBugFix` campaign, the same one the Delphes signal samples were generated from.

sample	events	size	note
signal $\kappa_\lambda = 0, 1, 5$	3×5.0M	7.3 GB	one `kl` column per event
$t\bar{t}$	298.7M	160.2 GB	1240/1244 shards (Sec. 3.4)
total	313.7M	167.5 GB	33 parquet files

Table 1: Merged ntuples on `/ceph`. Produced by a 1394-job HTCondor campaign, 22.4 min/job, 22.0 GB mean / 26.7 GB peak per job.

The κ_λ value is recorded per event. It is encoded only in the input path and the ntuple schema had no field for it, so a merged signal sample would have been unusable for NSBI — the method is entirely about ratios *between* κ_λ hypotheses. It is now recovered from the production plan at merge time and written as a column.

1.2 Tuning

The τ_h energy correction is the one substantive change since the last note. A multiplicative median scale factor cannot reshape a distribution, and the Delphes τ response carries a one-sided tail that CMS does not have (Fig. 1): at gen $p_T = 20$–30 GeV one Delphes τ in ten had more than twice the median response, against 1.08 on CMS. The correction now samples the anchor response *distribution* per object, which reproduces the CMS response by construction and fixes scale and width together. m_{vis}, $\tau_1\, p_T$, $\tau_2\, p_T$ and p_T^{miss} all overlay CMS after this change.

quantity	status	how
b-tag efficiency (b, c, light)	✓ tuned	stochastic re-tag from `Jet.Flavor` at anchor UParT-AK4 Medium
b-jet energy *scale*	✓ tuned	inverted to unity against `GenJet` (self-anchored), 0.965–0.989
b-jet energy *resolution*	× tuned	not corrected at all — Sec. 3.3
τ_h ID efficiency	✓ tuned	anchor `GenVisTau`→DeepTau v2.5 Medium; includes HPS reco inefficiency
τ_h energy response	✓ tuned	*distribution* resampled from anchor quantiles, not a median ratio
τ_h visible mass	✓ tuned	sampled per jet from anchor per-p_T quantiles
p_T^{miss} resolution	✓ tuned	Gaussian smearing binned in jet H_T
lepton reco efficiency	✓ tuned	per-p_T SFs, now 0.96–1.12 (e), 1.03–1.07 (μ)
di-τ mass estimator	✓ identical	same covariance-free FastMTT on both sides

Table 2: Object status. Every map is derived on the CMS anchor and applied downstream; none needs a card change.

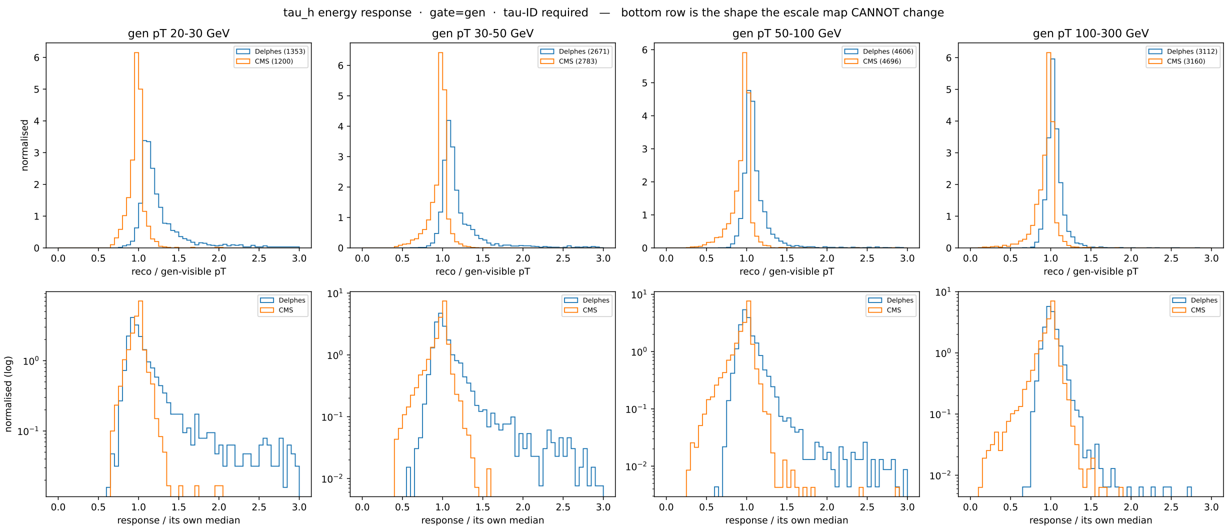

Figure 1: τ_h energy response, Delphes vs the CMS anchor. The Delphes tail above the median is what a multiplicative correction cannot remove and what motivated resampling the distribution instead.

2 Validation against CMS

2.1 The lepton definition — a defect found and fixed

The CMS τ-candidate pool was being built from the raw NanoAOD `Electron` and `Muon` collections, with *no* ID and no isolation requirement. Those collections are loose enough to contain jet fakes that no analysis would use, whereas a Delphes lepton has already passed the card's efficiency parameterisation, which is effectively an ID. The two pools therefore did not mean the same thing, and the consequences were not cosmetic:

- CMS acquired candidates Delphes cannot produce — 49 950 selected events against Delphes' 30 491 at $\kappa_\lambda = 0$, a 1.6× excess read at the time as a Delphes acceptance deficit.
- A fake lepton paired with a real τ is back-to-back, so the CMS $\Delta R_{\tau\tau}$ and ΔR_{bb} distributions grew a spurious second peak at $\Delta R \simeq \pi$, plus $m_{\tau\tau}$ and m_{vis} tails and a flattened $\cos\theta^*$.

- The same collections feed `lepton_efficiency`, which sets the `electron_sf` / `muon_sf` anchor targets. The scale factors were therefore tuned toward loose-reco efficiency rather than toward analysis-quality leptons.

The cut is now applied in the NanoAOD *reader*, so the tuning anchor and the validation overlay cannot use different lepton definitions again, and a configured-but-missing ID branch raises rather than silently reverting to the loose collection. Working points are `Electron_mvaIso_WP80`, `Muon_tightId` and `Muon_pfRelIso04_all` < 0.15; **these should be checked against the CMS $b\bar{b}\tau\tau$ baseline before anything is quoted.** After the fix the lepton fractions match (30% Delphes vs 33% CMS) and the scale factors sit near unity, which is itself the evidence that the two definitions now agree — against the loose collection they had to make up a ~20% gap.

2.2 Result

✓ agree	m_{HH} (the κ_λ observable, at $\kappa_\lambda = 0, 1, 5$); m_{bb} peak; p_T^{HH}; ΔR_{bb}; $\Delta\phi_{HH}$; p_T^{H1}, p_T^{H2}; $\cos\theta^*$; $\tau_1\, p_T$, $\tau_2\, p_T$; p_T^{miss}
× disagree	$\Delta R_{\tau\tau}$ (Delphes excess at 2.5–3.2), and the $m_{\tau\tau}$ / m_{vis} high tails that follow from it; m_{bb} *width*

Table 3: NSBI inputs, semi-leptonic channel, 200k events per κ_λ point against the `PowhegBugFix` anchor including its `ext1` extension. Selected yields Delphes/CMS: 30 491/28 181 ($\kappa_\lambda = 0$), 32 783/31 442 ($\kappa_\lambda = 1$), 24 218/20 443 ($\kappa_\lambda = 5$).

2.3 The $t\bar{t}$ background

The $t\bar{t}$ maps (`maps_ttbar_v1.json`) were applied to every $t\bar{t}$ event in the production but had never been compared back against the anchor they came from. They now have been, on `TTto2L2Nu` with the same selection on both sides, and the correction works: the stock card over-produces τ_h candidates by 1.68× (42 293 against CMS' 25 241 in the same 200k events) and the re-tag brings that to 1.22×. Selected yield moves from 0.70 to 0.89 of CMS.

All ten NSBI features agree in shape both before and after — which is the part to read carefully. The overlays are density-normalised, so a 1.68× error in τ_h *rate* is invisible in them. Shape agreement is not evidence that the τ_h object is right, and the 22% residual after tuning appears in no panel.

3 Open items

3.1 $\Delta R_{\tau\tau}$: too many wide-angle τ pairs

The one physics residual. Delphes has an excess at $\Delta R_{\tau\tau} \simeq 2.5$–$3.2$, and the $m_{\tau\tau}$ and m_{vis} high tails are probably the same events counted twice — but that is an inference, not a demonstration, so the argument is worth stating. For legs whose masses are negligible against their energies, $m_{\text{vis}}^2 = 2\,p_{T1}p_{T2}\,(\cosh\Delta\eta - \cos\Delta\phi)$ exactly, and both angular terms grow monotonically as the pair separates; $m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1x_2}$ inherits it. A wide pair is therefore a heavy pair — though m_{vis} is monotonic in $\Delta\eta$ and $\Delta\phi$ separately and is *not* a function of ΔR alone, so "wide-angle" rather than "back-to-back" is what ΔR measures. The converse does not follow by itself: a heavy tail could equally come from harder τ, since m_{vis} also grows with $p_{T1}p_{T2}$. What excludes that here is that $\tau_1\, p_T$ and $\tau_2\, p_T$ both overlay CMS, leaving the angular term as the only place the excess mass can come from. The direct test — cutting $\Delta R_{\tau\tau} < 2.5$ and asking whether the m_{vis} and $m_{\tau\tau}$ tails go with it — has not been run; if the tails survive, these are two problems and not one. This matters for the feature set: dropping $\Delta R_{\tau\tau}$ from the NSBI inputs would remove the most visible symptom while leaving the identical artefact in $m_{\tau\tau}$ and m_{vis}, which we certainly keep. The cause has to be found rather than worked around.

The leading hypothesis is τ_h mistag: a fake τ from the opposite hemisphere pairs with a real one at $\Delta R \simeq \pi$. This is directly testable — the overlay can split each side into gen-matched and fake τ pairs and report the fake fraction — and has not yet been run at full statistics. The $t\bar{t}$ overlay (Sec. 2.3) now supplies a first measurement of the size of the effect: the stock card makes $1.68\times$ too many τ_h candidates, so there is a real fake excess to account for. It is also actionable if confirmed: the `tau_fake_response` map has its top three p_T bins below the 200-entry threshold and is borrowing them from neighbours, so the high-p_T fake response is extrapolated rather than measured.

3.2 FastMTT returns no solution for most events

Found while preparing the CROWN-matched ntuples, and the more serious of the two new items. $m_{\tau\tau}$, m_{HH} and p_T^{HH} taken from FastMTT are `NaN` on 69% of signal and 80% of $t\bar{t}$ events, while every other branch on those rows is finite.

The mechanism is exact rather than statistical. The hadronic decay weight is non-zero only for $x \geq (m_{\text{vis}}/m_\tau)^2$, with $m_\tau = 1.777\,\text{GeV}$, so a leg whose visible mass exceeds m_τ has zero likelihood on the whole scan grid and the fit returns nothing. A Delphes τ_h *is* an AK4 jet and carries the jet mass — 5–15 GeV for a 30–50 GeV jet, three to eight times m_τ, where a real HPS τ_h never exceeds the a_1 mass. Failures therefore track the τ jet mass rather than being a random subset, which is why failing events have a higher visible $m_{\tau\tau}$ (92 vs 72 GeV).

Two consequences. The elliptical signal region evaluates to `NaN` on those events and rejects them, so it silently doubles as a FastMTT-success filter: a cut CMS quotes at 99% signal efficiency runs at roughly 31% here, on survivors biased in the very mass it selects on. And the failure rate differs between processes, so FastMTT does not cancel between signal and background as Sec. 5 assumed.

Mitigation is one line — clamp the τ_h mass to a physical τ visible mass before the fit — which removes the failures but leaves the jet-level energy scale untouched. The real fix is to rebuild the τ_h four-vector from the particle-flow constituents, already written to the Delphes files, which would also supply decay modes and a prong-summed τ charge.

3.3 m_{bb} width: the b-jet resolution is untuned

The b-jet correction inverts the response to unity against Delphes' *own* `GenJet`. That fixes the scale, does not touch the width, and never references CMS at all. The τ side, by contrast, now has its full response distribution resampled from the CMS anchor. m_{bb} is therefore the feature receiving the weaker of the two treatments, and its width does not match CMS.

The fix is the direct analogue of what worked for τ: measure the CMS b-jet response as a distribution and resample it. The machinery exists. The cost is the decision — it requires re-deriving the maps and re-running the full campaign ($\sim$10 h) — and it should be validated on a subset before that is spent. Note also that the card runs without pileup, so part of the CMS b-jet resolution has no counterpart in Delphes to correct.

3.4 Four lost $t\bar{t}$ shards

Four of 1244 $t\bar{t}$ shards (0.3%) could not be produced: the dCache redirector consistently routes those files to a pool node whose certificate does not cover the address it hands out. Bounded retries were added and confirm the failure is deterministic, not transient, so this is a site issue to report rather than a pipeline defect. Shards group input files by size with no relation to physics, so the loss costs MC statistics and introduces no bias; the shortfall is recorded in the merged manifest rather than hidden.

4 Decisions and next steps

1. **Confirm the lepton working points** against the CMS $b\bar{b}\tau\tau$ baseline. Everything downstream follows from them, and the current values are conventional choices rather than the analysis's.
2. **Clamp the τ_h mass in FastMTT** (Sec. 3.2). One line, and it unblocks every $m_{\tau\tau}$-based quantity; until it is in, use the visible mass.
3. **Run the gen-matched/fake split at full statistics** to test whether the $\Delta R_{\tau\tau}$ excess is τ mistag. Cheap, and it decides whether Sec. 3.1 is a fixable map problem or something deeper.
4. **Decide whether to tune the b-jet resolution** (Sec. 3.3). This is the only item costing a re-production, so it should be validated on a subset first.
5. **Then train.** The ntuples are ready and κ_λ-labelled; the feature set should be settled by measured κ_λ separation and by training with and without candidate features, not by inspecting one-dimensional marginals.

5 Known limitations

Accepted by design. FastMTT is covariance-free and is run identically on both sides. The cancellation argument made here previously no longer holds as stated: Sec. 3.2 shows the fit fails at different rates for signal and $t\bar{t}$, so the failures do not divide out. Until the mass clamp is in, $m_{\tau\tau}$ from FastMTT should be treated as unreliable and the visible mass used in its place. The card runs without pileup, so jet energy resolution is untuned and pileup jets cannot exist; neither is fixable downstream. p_T^{miss} smearing is now binned in jet H_T but remains isotropic where the real resolution is anisotropic.

A scope bound, not a defect. `tau_mistag` is measured on the b-rich signal anchor and fake rates are strongly jet-flavour dependent, so it is not valid for fake-dominated backgrounds (QCD, W+jets). The present signal study is unaffected; a background-heavy analysis would need those measured separately. There is likewise no CMS anchor for DY, so DY cannot currently be tuned.

Untested — risks, not accepted limitations. The sampled τ_h mass is drawn independently of the τ energy and of the partner leg, whereas the true visible mass correlates with decay mode; size unknown. Every map is flat-extrapolated above 300 GeV. That is probably benign here — only $\sim$2.8% of b-jets exceed it, the card's b-tag formula tends to a constant so flat is the right asymptote, and κ_λ discrimination lives near threshold ($m_{HH} \simeq 250$–400 GeV, where the triangle's s-channel propagator makes the interference most κ_λ-sensitive) rather than in the box-dominated tail. A boosted or high-m_{HH} category would change that, by making the extrapolated region the signal region.